

ECON1010

Introductory Microeconomics

Tutorial 6 | Perfectly Competitive Supply



This Week in ECON1010

CML 3 is due **next week** 4.00pm April 21

CLEAR-JE 3 is due **next week** 4.00pm April 23

Assessment

85% **7**

75% **6**

65% **5**

50% **4**

47% **3**

30% **2**

0% **1**

CMLs (25%)

- Computer Managed Learning
- Each worth 5%
- Best 5/6 count towards final grade

CLEAR-JE (20%)

- Contextualised Learning of Economics via Authentic Reflection using Journal Entries
- Each worth 10%
- Best 2/3 count towards final grade

Final Exam (55%)

Tools and Resources

I need HELP...

1. **Ed Discussion Board**
(Blackboard)
2. **Consultation** (8.00am
Mon, 39-125A)
3. **Staff email**
econ1010@uq.edu.au

I REALLY need HELP...

- <https://my.uq.edu.au/contact/student-central>

ECON1010 Consult Timetable					
Semester 1 2026					
Time	Mon	Tue	Wed	Thu	Fri
8:00-8:30	CONSULT - SAM 39-125A				
8:30-9:00					
9:00-9:30					
9:30-10:00					
10:00-10:30				CONSULT - LUCAS 39-125A	
10:30-11:00					
11:00-11:30	CONSULT - GABE 39-125A	CONSULT - BORIS 39-125A	CONSULT - IKMAL 39-125A	CONSULT - NIK 39-125A	CONSULT - CARL 39-641
11:30-12:00					
12:00-12:30					
12:30-13:00					
13:00-13:30					LECTURE
13:30-14:00		CONSULT - DYLAN 39-125A			
14:00-14:30				CONSULT - GUANCHU 39-125A	
14:30-15:00					
15:00-15:30					
15:30-16:00					
16:00-16:30			CONSULT - KIRTAN 39-125A	CONSULT - NANDAN 39-125A	
16:30-17:00					
17:00-17:30					

Student Central

Need help? We've got you covered.

Contact us

Monday to Friday

- Email
- 1300 275 870 (8.30am-5pm)
- Building 42, St Lucia
- Building 8101A, Gatton

Chat – 1:51 (1 Minute) wait

Emergency help

For immediate risk:

UQ Campus Security
07 3365 3333 (24/7)

Off-campus emergency
000 (24/7)

Crisis support

For urgent mental health support:

UQ Counselling and Crisis Line
1300 851 998 (24/7)

Text a Crisis Counsellor
0488 884 115 (4.30pm-8am)

Sexual Misconduct Support Unit (SMSU)

Perfectly Competitive Supply

Market Structures

Increasing Competition



**Perfect
Competition**

**Monopolistic
Competition**

Oligopoly

Monopoly

Increasing Concentration



Question 1ab

a) In your own words, what do you think is:

- i) An essential feature of a Perfectly Competitive Market?
- ii) The most important cost to consider?
- iii) Loss minimising?
- iv) The shutdown point?
- v) The difference between the short and long run?

b) On a graph, draw the demand curve an individual firm faces when supplying a market under perfect competition.

Discuss with your table group, choose one person to share

Remember – the individual supplier has:

- *Total Fixed Costs (TFC)*
- *Total Variable Costs (TVC)*
- *Total Costs (TC)*
- *Average Total Costs (ATC)*
- *Average Fixed Costs (AFC)*
- *Average Variable Costs (AVC),*
- *Marginal Cost (MC).*



Question 1a

i) An essential feature of a Perfectly Competitive Market?

- **Many buyers and many sellers** – many sellers allow consumers to change supplier if one raises their price. Many consumers prevent the consumer having too much power.
- **Similar products exist** (homogeneous products) – substitute goods allow consumers to change products if prices rise.
- **Easy entry and exit from the industry** – low entry costs, eg doesn't prevent new sellers from entering the market because it's too expensive, and no barriers, eg medical regulations restrict the number of doctors available and the ability of companies to create and sell drugs.
- **Resources are perfectly mobile** – free movement of goods and capital allow easy entry and exit from the market.
- **Perfect knowledge and information** – the buyers or sellers cannot 'cheat' or trick the other into paying an unfair price as both know the item/product true value.

ii) The most important cost to consider?

All of the costs are useful in different ways.

- **The marginal cost** is important as it determines how much quantity a supplier will sell.
- **The average variable cost** is important as its minimum is the 'shut down' point (explained next).

Question 1a

iii) Loss minimising?

Loss minimising is when a supplier is not making a profit overall but continues selling goods to reduce losses. This only occurs when the price is above average variable costs and below average total costs. The quantity produced should be where price = marginal cost. This will minimise the supplier's losses (by recovering all the variable costs but only part of the money already spent on fixed costs).

iv) The shutdown point?

The shutdown point is where it's more beneficial for the supplier to shut down operations and leave the market. This is where the price is below the minimum average variable costs, which means the supplier loses more money with every unit sold.

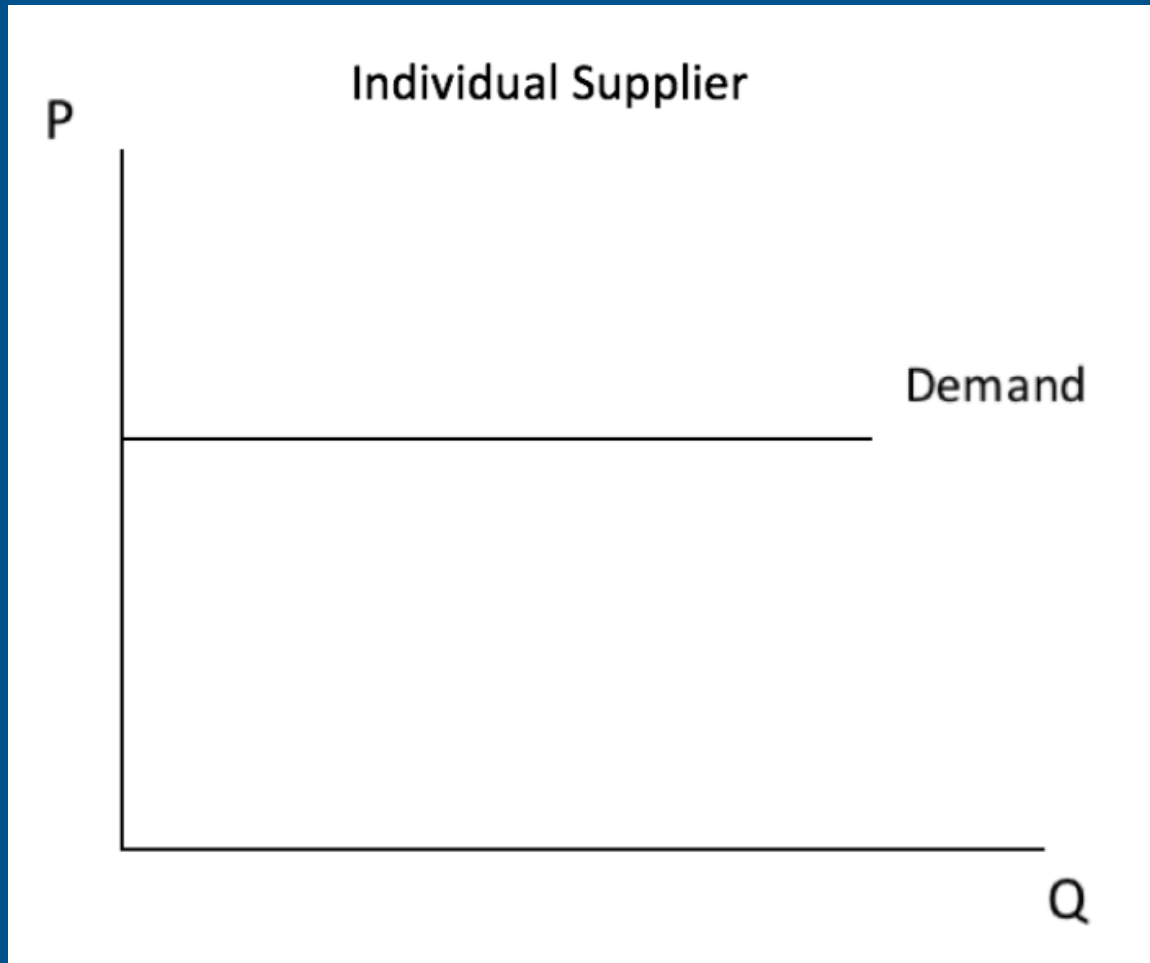
v) The difference between the short and long run?

The short run is the time period where some inputs are variable but others are fixed (such as employees/factories, as it takes time to find, hire, police check and train new employees or finalise the legal paperwork on a new factory purchase), which limits many essential features of a perfectly competitive market.

The long run, however, is the time period where all inputs can change, allowing greater flexibility in the production of goods.

Question 1b

On a graph, draw the demand curve an individual firm faces when supplying a market under perfect competition.



As the individual supplier/firm is competing against many other sellers in a perfectly competitive market, consumers will pay the same price for any quantity that any individual firm wishes to sell. If one supplier decides to increase their price above all other suppliers, consumers will simply purchase from one of the many other suppliers at the lower price. This is why suppliers in perfect competition are called “price takers” and all sell at the same price.

Question 1c

c) Complete the following table.

$$TC = FC + VC$$

$$VC = TC - FC$$

$$AVC = VC/Q$$

$$MC = \Delta TC / \Delta Q$$

$$TC = ATC * Q$$

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0	24						
1							16
2			50				
3	108						
4							52
5					39.2		
6				47			



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6		24		47			

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1	40	24					16
2		24	50				
3	108	24					
4		24					52
5		24			39.2		
6		24		47			

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1	40	24	16				16
2		24	50				
3	108	24					
4		24					52
5		24			39.2		
6		24		47			

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1	40	24	16	40	16	24	16
2		24	50				
3	108	24					
4		24					52
5		24			39.2		
6		24		47			

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1	40	24	16	40	16	24	16
2	74	24	50				
3	108	24					
4		24					52
5		24			39.2		
6		24		47			

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1	40	24	16	40	16	24	16
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3	108	24					
4		24					52
5		24			39.2		
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4	160	24					52
5		24			39.2		
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4	160	24	136				52
5		24			39.2		
6		24		47			

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2	74	24	50	37	25	12	34
3	108	24	84	36	28	8	34
4	160	24	136	40	34	6	52
5	220	24	196		39.2		
6		24		47			

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5	220	24	196		39.2		60
6		24		47			

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3	108	24	84	36	28	8	34
4	160	24	136	40	34	6	52
5	220	24	196	44	39.2	4.8	60
6		24		47			

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5	220	24	196	44	39.2	4.8	60
6	282	24		47			

Question 1c

c) Complete the following table.

$$TC = FC + VC$$

$$VC = TC - FC$$

$$AVC = VC/Q$$

$$MC = \Delta TC / \Delta Q$$

$$TC = ATC * Q$$

$$VC = AVC * Q$$

$$AVC = ATC - AFC$$

$$FC = TC - VC$$

$$ATC = AFC + AVC$$

$$AFC = FC/Q$$

$$FC = AFC * Q$$

$$ATC = TC/Q$$

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Quantity (units)	Total Cost (\$)	Fixed Cost (\$)	Variable Cost (\$)	Average Total Cost (\$/unit)	Average Variable Cost (\$/unit)	Average Fixed Cost (\$/unit)	Marginal Cost (\$/unit)
0	24	24	0	-	-	-	-
1	40	24	16	40	16	24	16
2	74	24	50	37	25	12	34
3	108	24	84	36	28	8	34
4	160	24	136	40	34	6	52
5	220	24	196	44	39.2	4.8	60
6	282	24	258	47			

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6	282	24	258	47	43	4	62

Question 2a

a) You did really well in your ECON1010 final exam and decide to offer private online microeconomics tutoring. You recognise you will need to consider the following costs:

1. Fixed costs: laptop and software.
2. Variable costs: tutoring online platform (pay per minute) and internet usage.
3. Opportunity cost: your current job.

What assumptions are necessary to make this a perfectly competitive market?

For this to be a perfectly competitive market there must be:

- **Many other online microeconomics tutors and students.**
- **Easy entry/exist – low fixed costs.**
- **Perfect knowledge and information – students can view other tutors' prices and verify the quality of their microeconomics tutoring skills.**
- **Homogenous good – all microeconomic tutors are providing the same tuition level or quality.**
- **Perfect mobility of factors of production – online classes and no contract or visa issues preventing workers from changing jobs or hours worked.**

Question 2a

- a) Now consider online microeconomics tutoring is a perfectly competitive market. When would an economist advise you to:
- i) Shut down your online tutoring operation.
 - ii) Change your tutoring price.
 - ii) Reduce hours of tutoring to earn greater profits or smaller losses.
 - iv) Increase hours of tutoring to earn greater profits or smaller losses.
 - v) Leave your hours of tutoring decision unchanged.



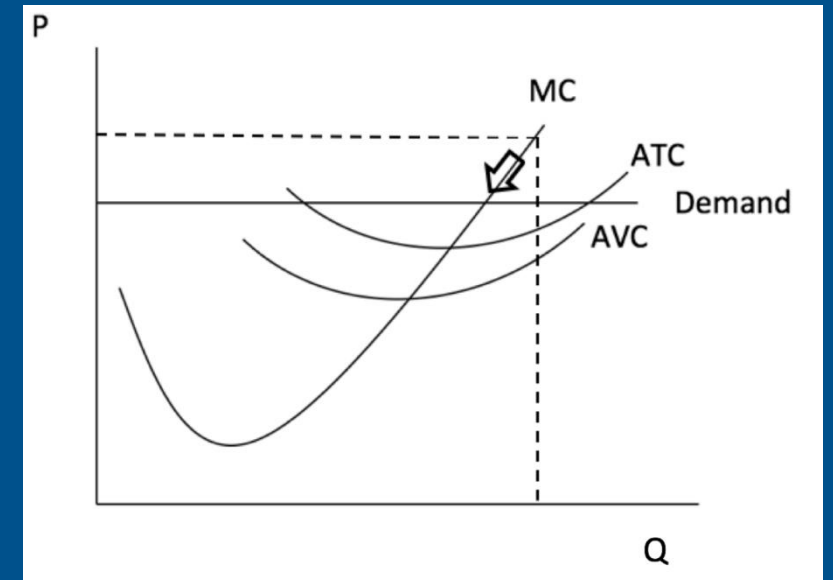
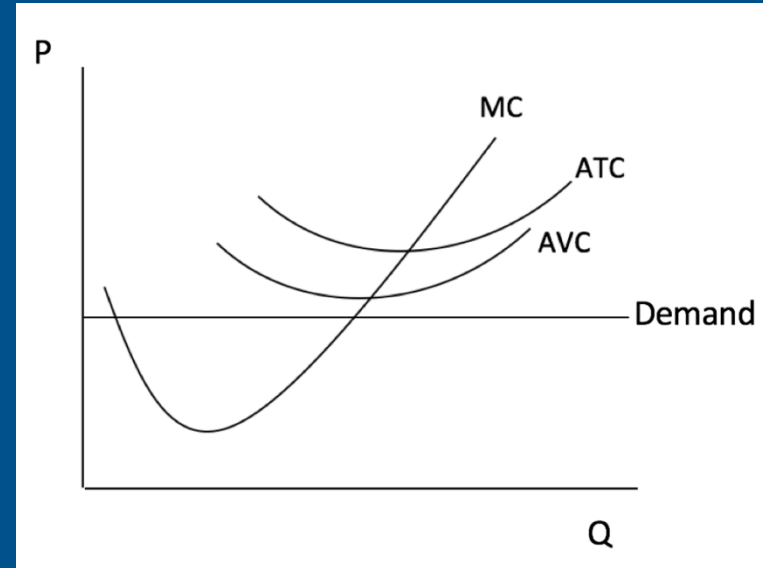
Question 2a

- i) *Shut down your online tutoring operation.*

When price is less than the minimum of the average variable cost.

- ii) *Change your tutoring price.*

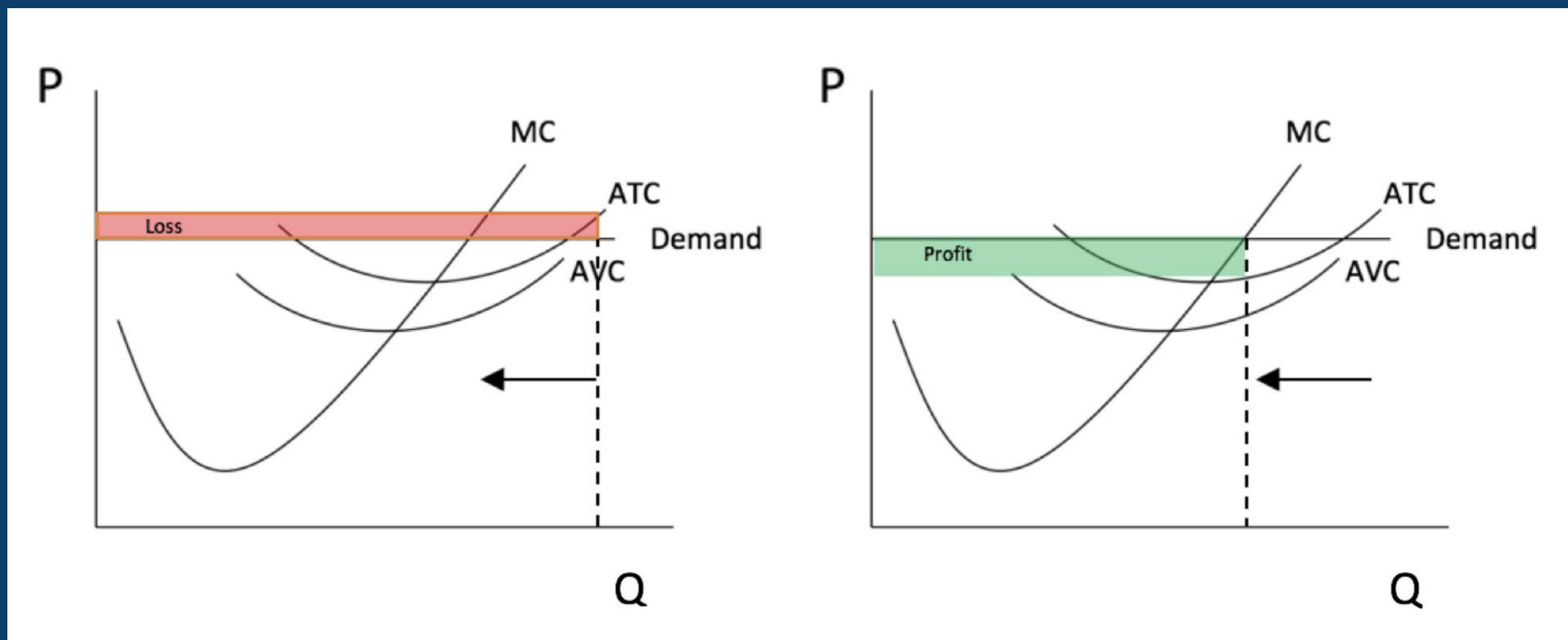
When price is not equal to demand.
If your tutoring price is above the market price, there will be no quantity demanded at that price.
Price would need to fall and hours of tutoring set where price = marginal cost.
If the individual price is below the market price, you will receive less revenue (and less profit) than you otherwise could.



Question 2a

iii) Reduce hours of tutoring to earn greater profits or smaller losses.

When Q is larger than the intersection of MC and Demand (= price).

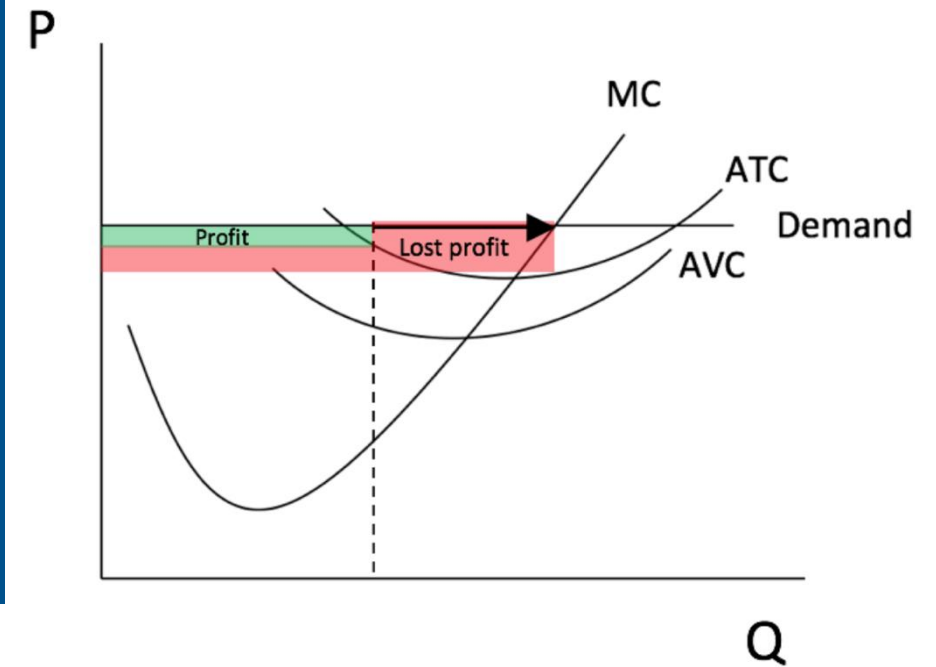


Profit-maximising output is where:
 $P = MC$

Question 2a

iv) Increase hours of tutoring to earn greater profits or smaller losses.

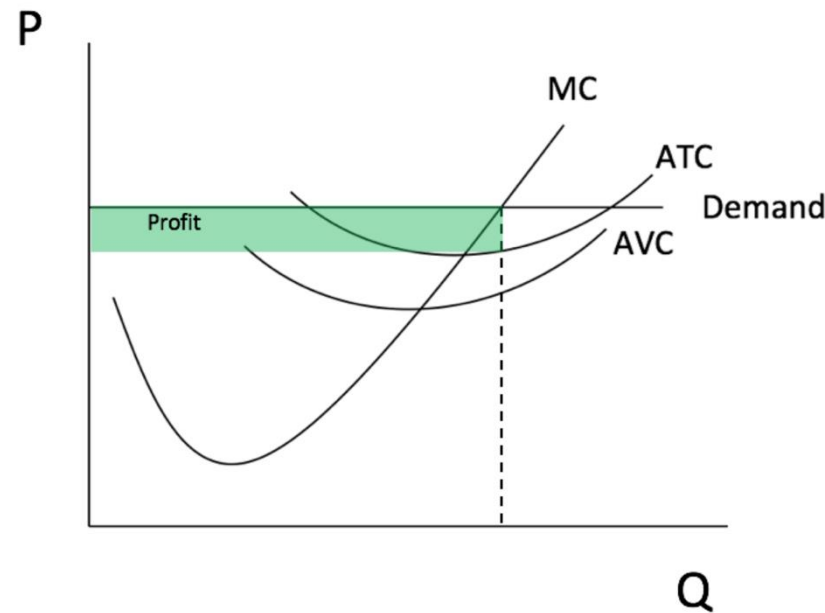
When quantity is less than the intersection of MC and Demand.



v) Leave your hours of tutoring decision unchanged.

When quantity is at the intersection of MC and Demand.

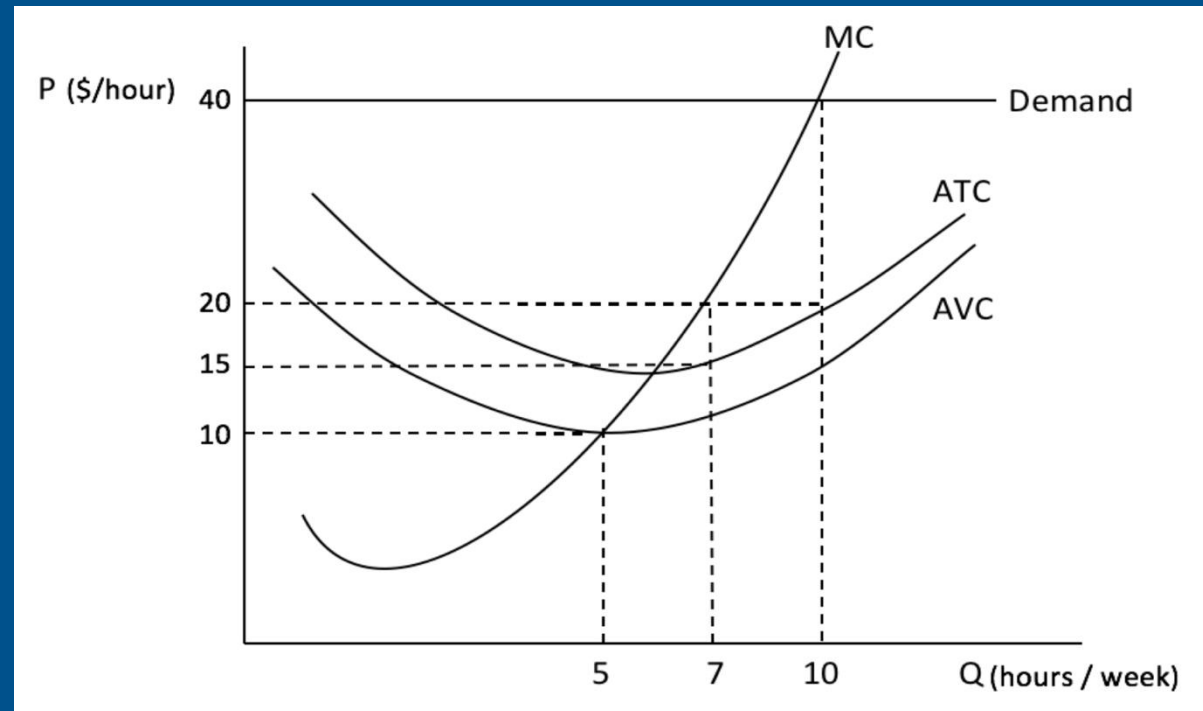
**Remember – profit-maximising output is where:
 $P = MC$**



Question 2b

The diagram below represents your online tutoring costs per week in the short run. As you calculate the answers for the questions below, clearly mark them on the diagram:

- What is your profit-maximising output (number of hours per week)?
- At the profit maximising output, what is your total revenue per week?
- At the profit maximising output, what are your total costs?
- At the profit maximising output, what is your profit (or loss)?
- At what price should you stop tutoring (shut down)?
- If the microeconomics online tutoring market price decreases from \$40/hour to \$20/hour, how many hours/week should you tutor to maximise profit or minimise losses?
- At this new output level (price = \$20/hour) do you earn more or less profit than in part iv)?



Question 2b

i) What is your profit-maximising output (number of hours per week)?

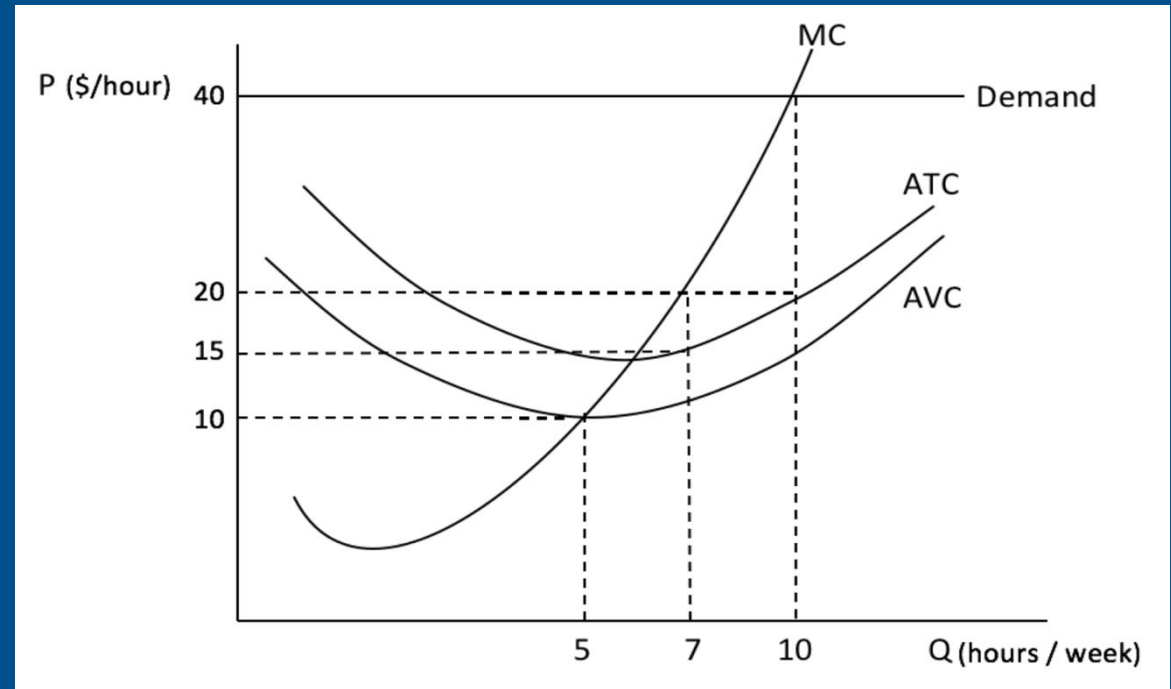
Profit maximising output is where $MC = \text{Demand}$ (Price). 10 hours/week.

ii) At the profit maximising output, what is your total revenue per week?

**Total Revenue = Price * quantity
= \$40/hour * 10 hours/week
= \$400/week**

iii) At the profit maximising output, what are your total costs?

**Total Costs = Average Variable Costs * quantity
= \$20/hour * 10 hours/week
= \$200/week**



Question 2b

iv) At the profit maximising output, what is your profit (or loss)?

$$\begin{aligned}\text{Profit} &= (\text{Marginal Cost} - \text{Average Total Cost}) * \text{quantity} \\ &= (\$40/\text{hour} - \$20/\text{hour}) * 10 \text{ hours/week} \\ &= \$200/\text{week}\end{aligned}$$

v) At what price should you stop tutoring (shut down)?

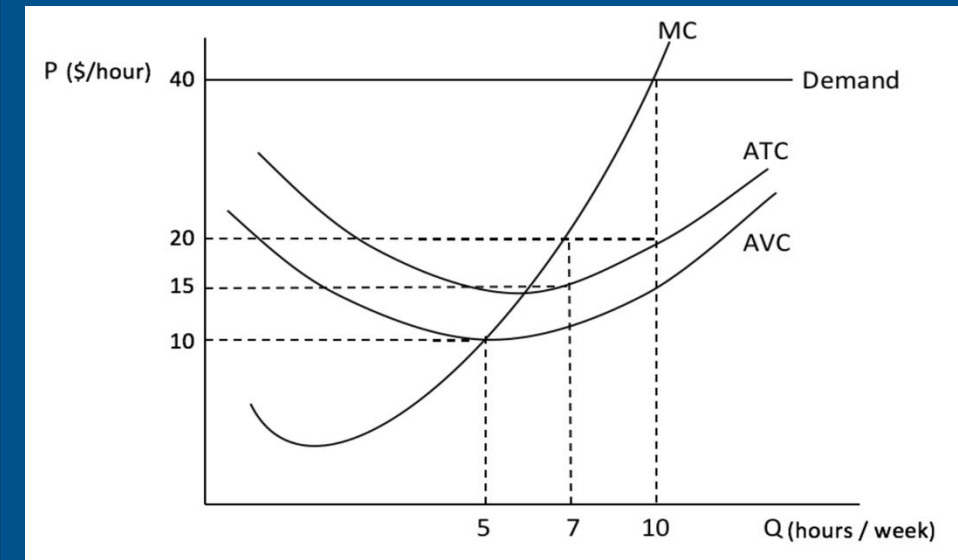
Stop tutoring when price is below \$10/hour, which is the minimum of AVC.

vi) If the microeconomics online tutoring market price decreases from \$40/hour to \$20/hour, how many hours/week should you tutor to maximise profit or minimise losses?

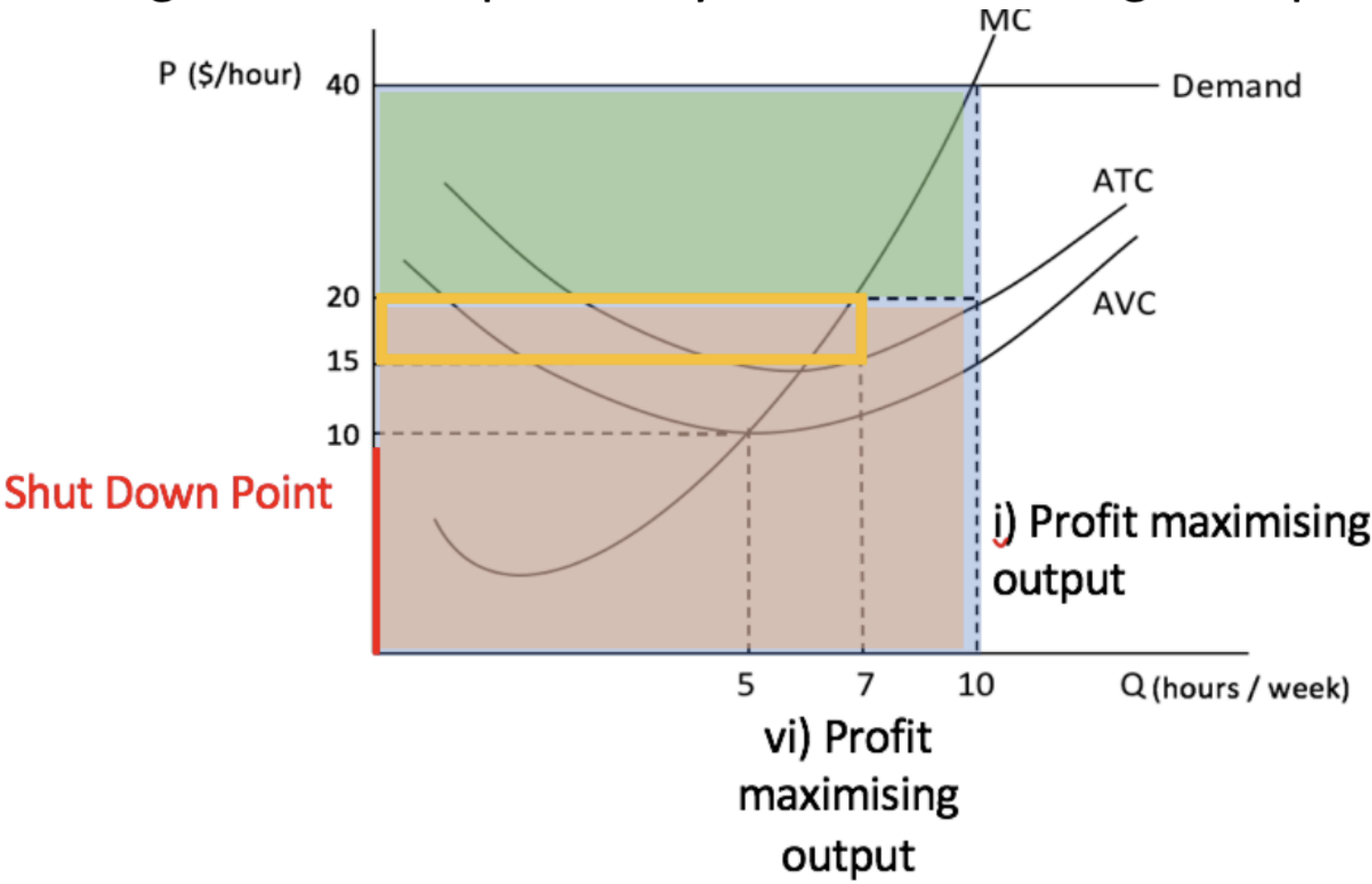
Should tutor 7 hours/week, as this is the new intersection between MC and Demand.

vii) At this new output level (price = \$20/hour) do you earn more or less profit than in part iv)?

$$\begin{aligned}\text{Profit} &= (\text{MC} - \text{ATC}) * Q \\ &= (\$20/\text{hour} - \$15/\text{hour}) * 7 \text{ hours/week} \\ &= \$35/\text{week profit. At } \$20/\text{hour, you will earn less profit than in part iv).}\end{aligned}$$



Question 2b



ii) Profit \$200 / week

vii) Profit \$35 / week

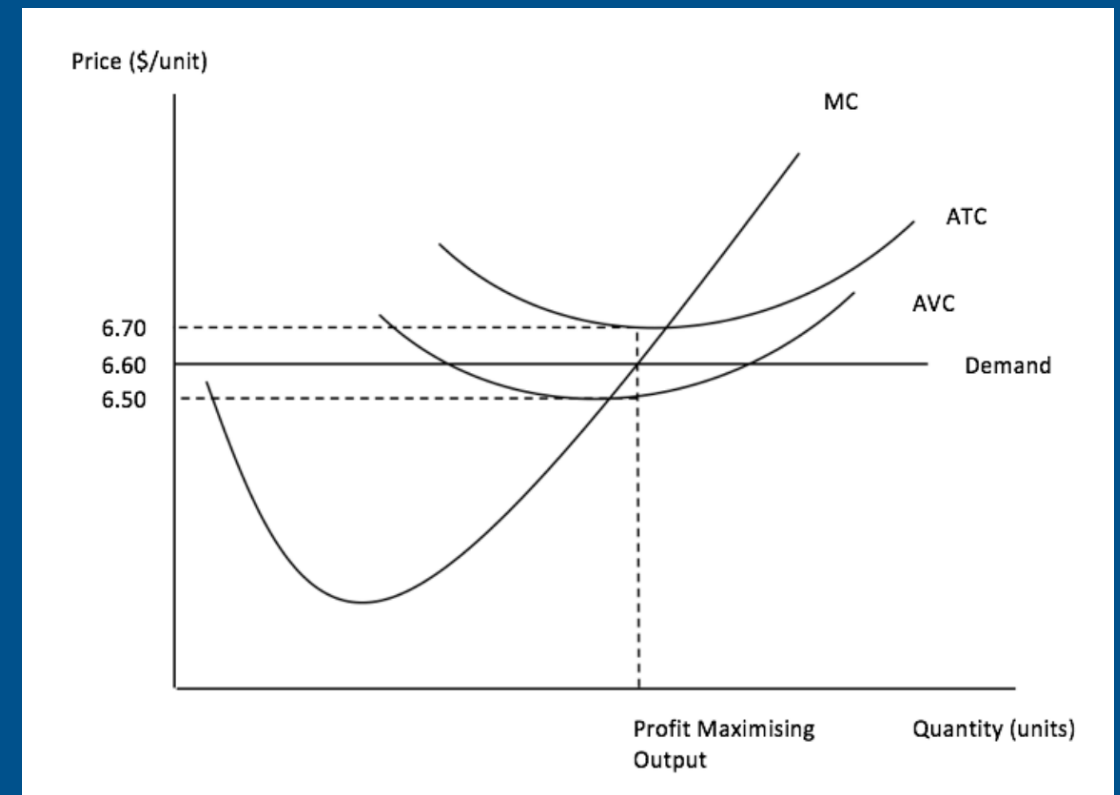
Total Costs \$200 / week

Total Revenue \$400 / week

Question 2c

A perfectly competitive firm is producing at the profit-maximising output level and finds its average variable cost (AVC) is **\$6.50/unit**, its average total cost (ATC) is **\$6.70/unit**, and it faces a market price (P) of **\$6.60/unit**. For each of the following statements, state whether the statement is true or false and explain why. Use the diagram below to justify your answer.

- i) The firm is earning an economic profit of 10 cents per unit.
- ii) The firm should absorb the 10 cent loss per unit and continue operating where price equals marginal cost.
- iii) The firm should shut down because it will lose less than if it continues to operate.
- iv) The firm should increase production to decrease the total cost per unit.



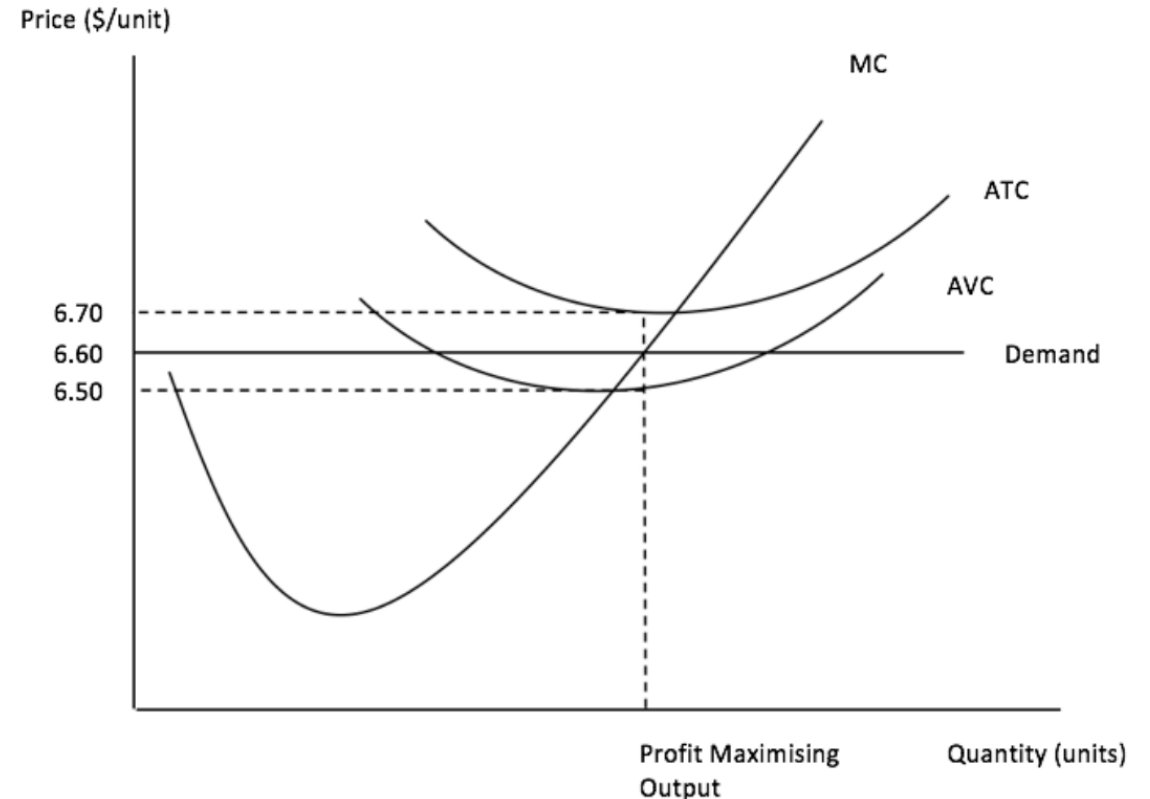
Question 2c

i) The firm is earning an economic profit of 10 cents per unit.

False. The firm is making a loss of 10 cents per unit (\$6.60/unit - \$6.70/unit).

ii) The firm should absorb the 10 cent loss per unit and continue operating where price equals marginal cost.

True. As the price remains above the average variable cost (\$6.50/unit), the firm will be loss minimising.



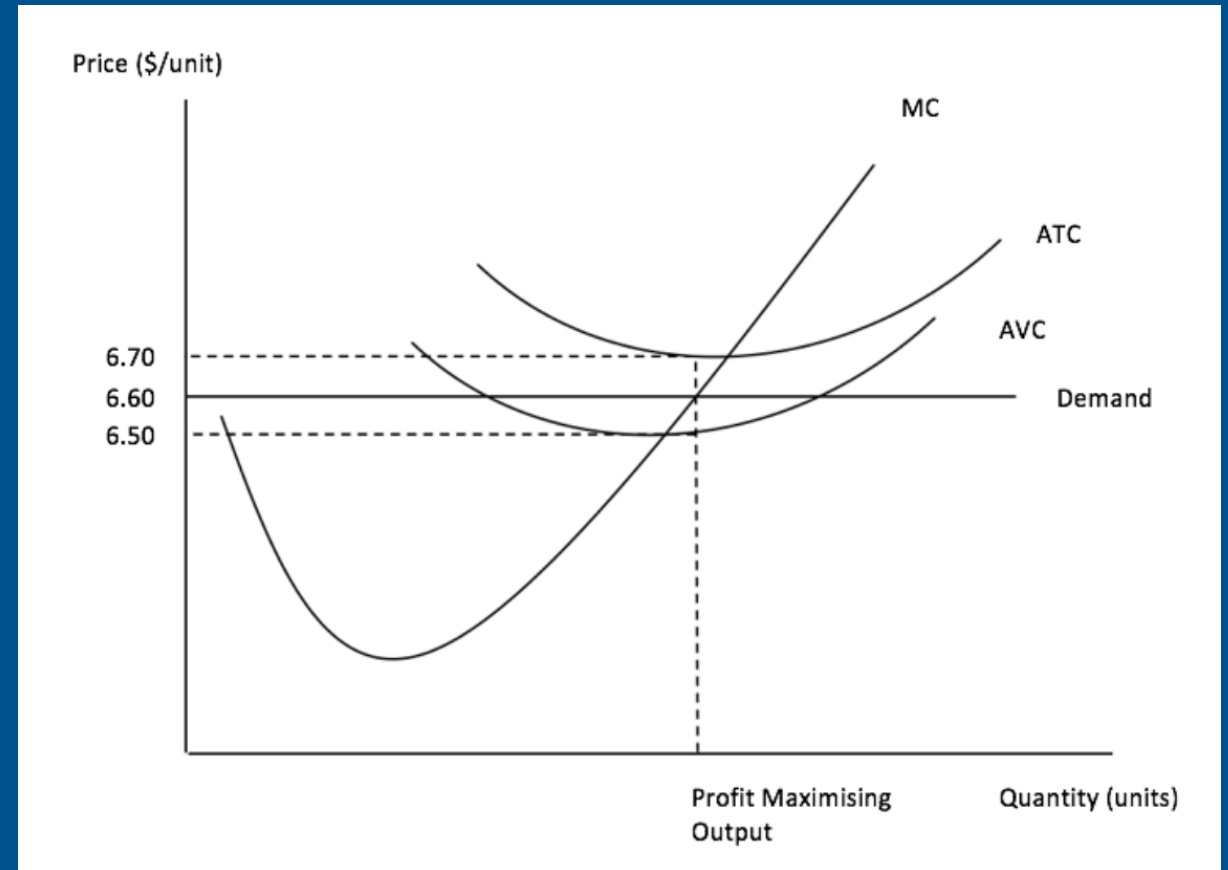
Question 2c

iii) The firm should shut down because it will lose less than if it continues to operate.

False. As the price is above the AVC it is recovering all the variable costs, along with some of the money spent on fixed costs.

iv) The firm should increase production to decrease the total cost per unit.

False. Even though average total cost may slightly decrease initially, it would soon begin to rise. Average total cost per unit would rise. Therefore increasing production will see average total cost per item continually increase above the constant selling price. The firm would make larger and larger losses as output continues to increase with price remaining constant.



Question 3

a) Reflect on your own personal situation:

- i) What is a perfectly competitive market that you interact with on a daily or weekly basis?
- ii) How many of the perfectly competitive market features does that market satisfy?



Thank You

Next week: The Quest for Profits

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